the eye, the random detection of individual photons can be seen. This is called *statistical noise*, and is encountered in all low-light imaging, such as military night vision systems. Chapter 25 will revisit this topic. Since rods cannot detect color, low-light vision is in black and white.

The cone receptors are specialized in distinguishing color, but can only operate when a reasonable amount of light is present. There are three types of cones in the eye: red sensitive, green sensitive, and blue sensitive. This results from their containing different *photopigments*, chemicals that absorbs different wavelengths (colors) of light. Figure 23-6 shows the wavelengths of light that trigger each of these three receptors. This is called **RGB encoding**, and is how color information leaves the eye through the optic nerve. The human perception of color is made more complicated by neural processing in the lower levels of the brain. The RGB encoding is converted into another encoding scheme, where colors are classified as: red *or* green, blue *or* yellow, and light *or* dark.

RGB encoding is an important limitation of human vision; the wavelengths that exist in the environment are lumped into only three broad categories. In comparison, specialized cameras can separate the optical spectrum into hundreds or thousands of individual colors. For example, these might be used to classify cells as cancerous or healthy, understand the physics of a distant star, or see camouflaged soldiers hiding in a forest. Why is the eye so limited in detecting color? Apparently, all humans need for survival is to find a *red* apple, among the *green* leaves, silhouetted against the *blue* sky.

Rods and cones are roughly 3 μm wide, and are closely packed over the entire 3 cm by 3 cm surface of the retina. This results in the retina being composed of an array of roughly 10,000 × 10,000 = 100 million receptors. In comparison, the optic nerve only has about one-million nerve fibers that connect to these cells. On the average, each optic nerve fiber is connected to roughly 100 light receptors through the connecting layer. In addition to consolidating information, the connecting layer enhances the image by sharpening edges and suppressing the illumination component of the scene. This biological image processing will be discussed in the next chapter.

Directly in the center of the retina is a small region called the **fovea** (Latin for *pit*), which is used for high resolution vision (see Fig. 23-4). The fovea is different from the remainder of the retina in several respects. First, the optic nerve and interconnecting layers are pushed to the side of the fovea, allowing the receptors to be more directly exposed to the incoming light. This results in the fovea appearing as a small depression in the retina. Second, only cones are located in the fovea, and they are more tightly packed that in the remainder of the retina. This absence of rods in the fovea explains why night vision is often better when looking to the *side* of an object, rather than directly at it. Third, each optic nerve fiber is influenced by only a few cones, proving good localization ability. The fovea is surprisingly small. At normal reading distance, the fovea only sees about a 1 mm diameter area, less than the size of a single letter! The resolution is equivalent to about a 20×20 grid of pixels within this region.